

WEEK 8

Remote Sensing Basics & Multi-Hazard Intuition

Identifying the Fingerprints of Disaster

ARIA v5.0 · The Matai'an Three-Act Auditor

Remote Sensing & Spatial Information Analysis

Prof. 蘇文瑞 · Department of Civil Engineering, National Taiwan University
National Science and Technology Center for Disaster Reduction (NCDR)

From Models to Witnesses

W3

WHERE
shelters

W4

HOW MUCH
terrain

W5

WHEN
rain

W6

WHAT-IF
Kriging

W7

CAN WE REACH?
network

W8

WHAT HAPPENED?
eyewitness

simulation → observation

THE 2025 CASE THAT BROKE HUALIEN

Jul 21: Typhoon Wipha's rainfall triggers a massive landslide in the upper Matai'an Creek, Wanrong Township. Debris dams the river.

64 days: A ~200 m deep barrier lake grows in full sunlight, on public Sentinel-2 imagery, freely downloadable.

Sep 23, 14:50: The lake breaches. 15.4 million m³ released in 30 minutes. Guangfu Township is buried. Highway 9 Matai'an bridge collapses. 18+ lives lost.

What is Remote Sensing?

Observing Earth without touching it.

PASSIVE SENSORS

Capture reflected sunlight

Optical · Sentinel-2 · Landsat

Daylight dependent

Blocked by clouds

ACTIVE SENSORS

Emit their own signal

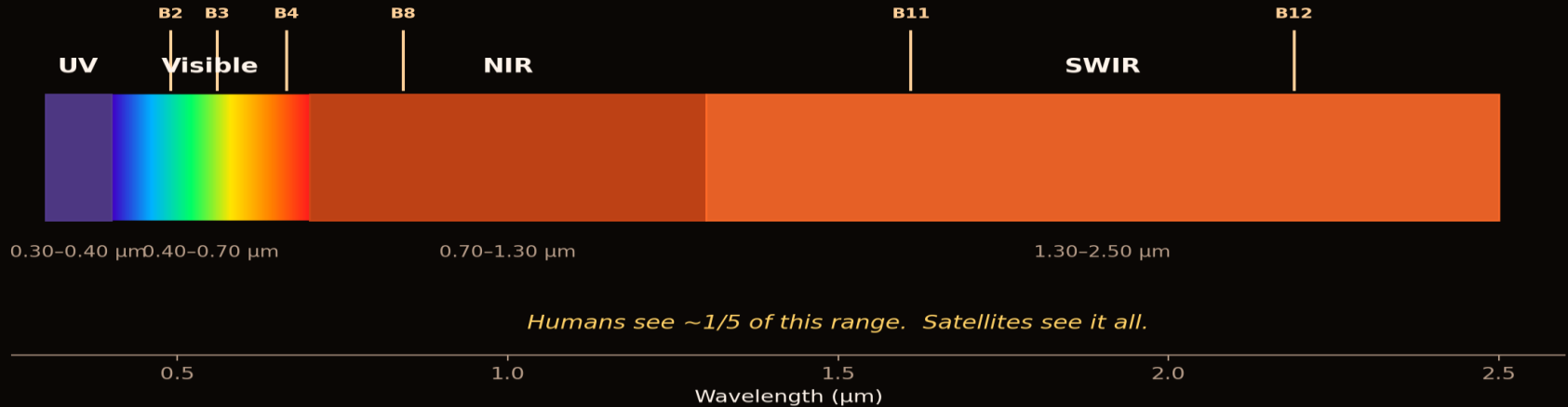
SAR · Sentinel-1 · Radar, Lidar

Day or night

Sees through clouds → Week 10

THE MATAI'AN LESSON: *Between Jul 21 and Sep 23, the barrier lake was visible on every cloud-free Sentinel-2 revisit. The question is never “do we have the data?” — it is “does anyone look?”*

The Electromagnetic Spectrum



VISIBLE

0.4–0.7 μm

What humans see

NIR

0.7–1.3 μm

Vegetation health

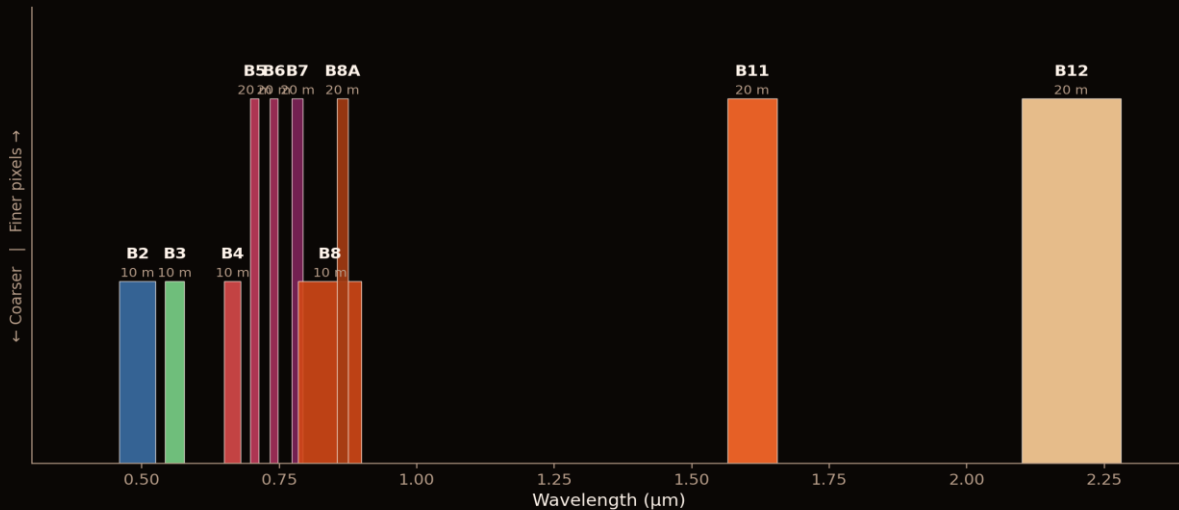
SWIR

1.3–2.5 μm

Moisture, burn, bare rock

Sentinel-2

Sentinel-2 Band Anatomy — Our Working Set



13

spectral bands

10 m

finest pixel

5 d

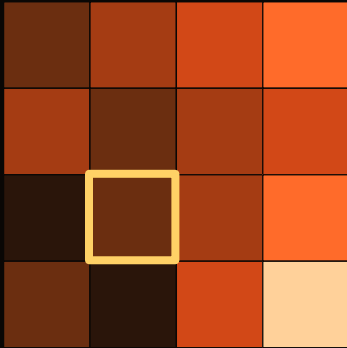
revisit time

0 \$

public, free

MATAI'AN COVERAGE: Between Jul 21 and Sep 23 2025, Sentinel-2 captured the barrier lake area ~12 times. You will use 3 of them today.

A Pixel is Not a Color



one pixel



SPECTRAL VECTOR

B2	blue	0.07	
B3	green	0.09	
B4	red	0.06	
B8	NIR	0.42	
B11	SWIR-1	0.21	
B12	SWIR-2	0.13	

DN vs SURFACE REFLECTANCE

Digital Number → raw integer counts

Surface reflectance → 0.0 – 1.0

Dimensionless, physical, comparable

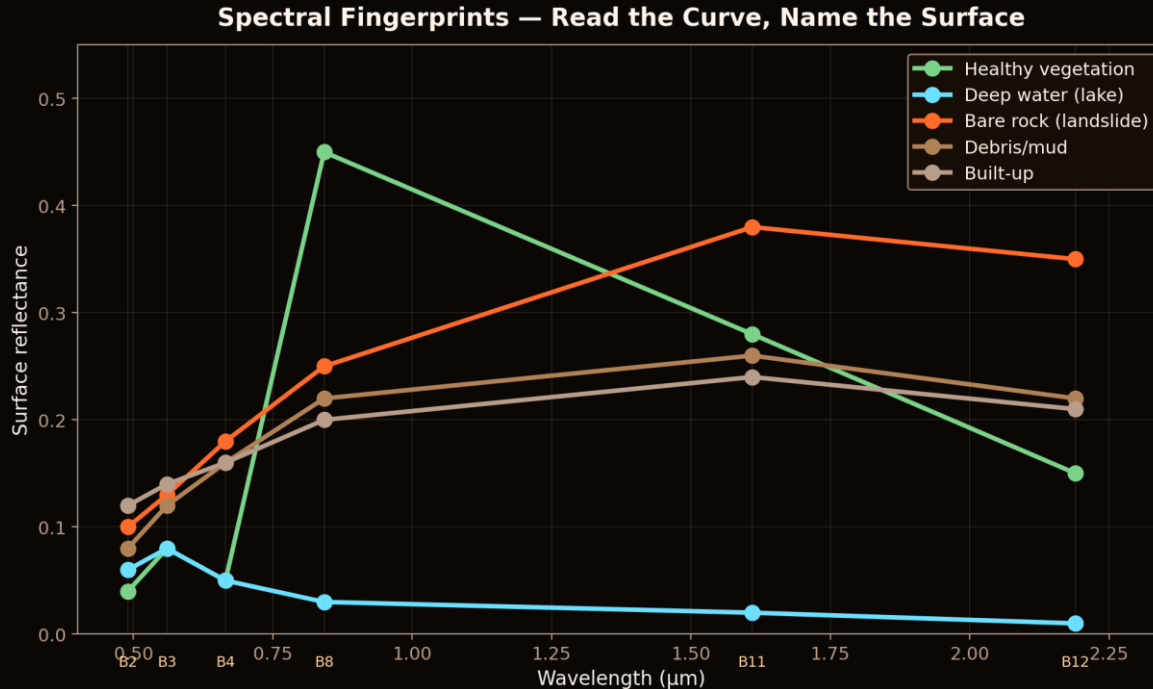
L1C vs L2A

L1C → Top-of-Atmosphere (TOA)

L2A → Bottom-of-Atmosphere (BOA)

Always L2A unless you have a very good reason.

Spectral Signatures — the Fingerprint



Water

high B3 · near-zero B8

Healthy veg.

low B4 · high B8

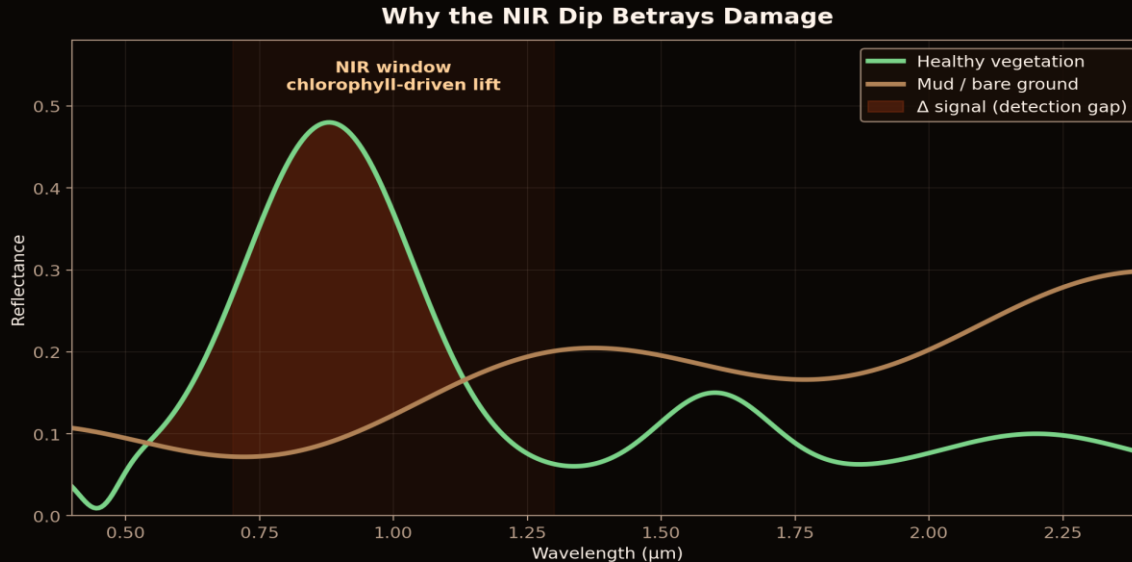
Bare rock

mid B4 · high B11/B12

Burned area

low B8 · high SWIR

Why Plants Reflect NIR



Chlorophyll

absorbs blue + red for photosynthesis

Spongy mesophyll

mirrors NIR — the bright signal

NIR drop

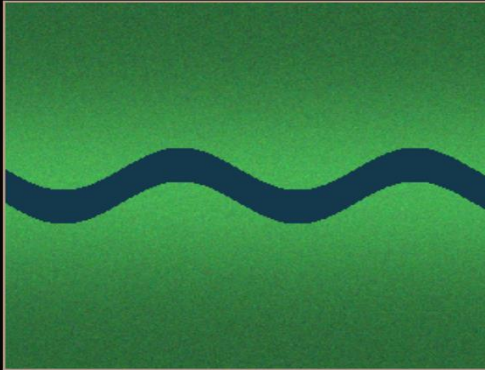
tissue is dead, submerged, or buried

CAPTAIN'S SHORTCUT: *Low NIR on a pixel that used to be green = something bad happened.*

Color Composites

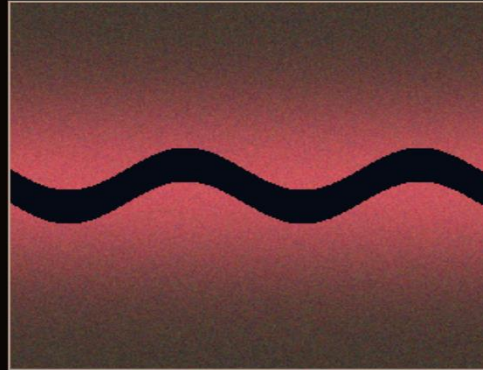
One Scene, Three Stories — Composite Strategy

True Color
B4·B3·B2



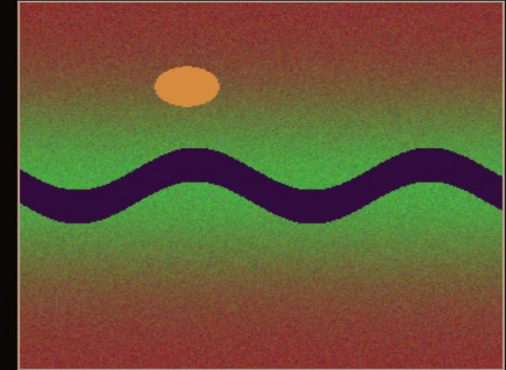
Vegetation = green, water = dark turquoise

False Color
B8·B4·B3



Vegetation = bright red

SWIR Composite
B12·B8·B4

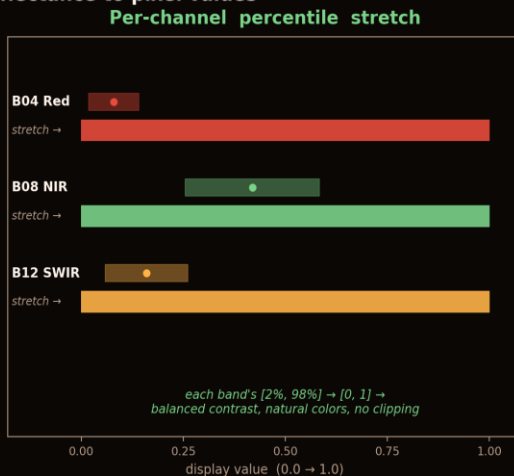
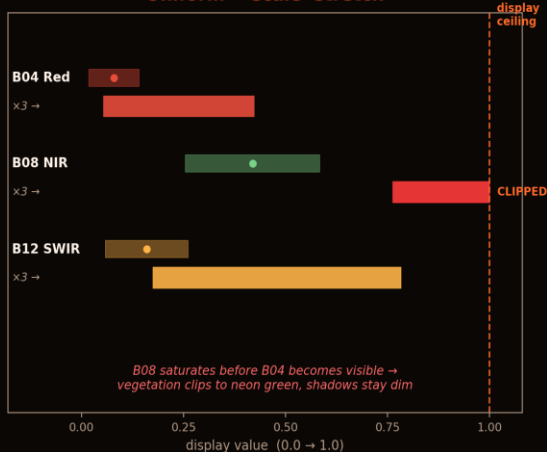


Bare rock = orange, moisture = blue

CAPTAIN'S STRATEGY: Always view at least two composites to cross-verify. One composite is a guess. Three composites is a diagnosis.

Display Stretching

Display stretching — from physical reflectance to pixel values



METHOD 1

```
np.clip(x * 3, 0, 1)
```

fast · one line · primitive

⚠ B08 saturates first

METHOD 2

per-band [2%, 98%] → [0, 1]

each band independent

✓ industry standard

CAPTAIN'S INSIGHT: Stretching is a *display* operation — it does NOT change the cube. All band math (NDWI, NDVI, NBR, BSI) must still use the raw reflectance values. A pretty composite is a *visualization*, not a measurement.

Spatial Resolution — Choose by the Question

10 m

B2 · B3 · B4 · B8

typical use:

Settlement · road · field · shelter · lake boundary

20 m

B5 · B6 · B7 · B11 · B12

typical use:

Vegetation health · landslide · moisture

60 m

B1 · B9 · B10

typical use:

Atmospheric correction only — not for surface

CAPTAIN'S RULE: *Don't use 60 m bands to assess bridges. Ever.*

TCI Quick-QA

TCI Quick-QA — the 10-second cloud test



WORKFLOW

- 1 STAC query per act
- 2 Sort by eo:cloud_cover
- 3 TCI-preview top 3
- 4 Skip only if cloud wall sits over your AOI

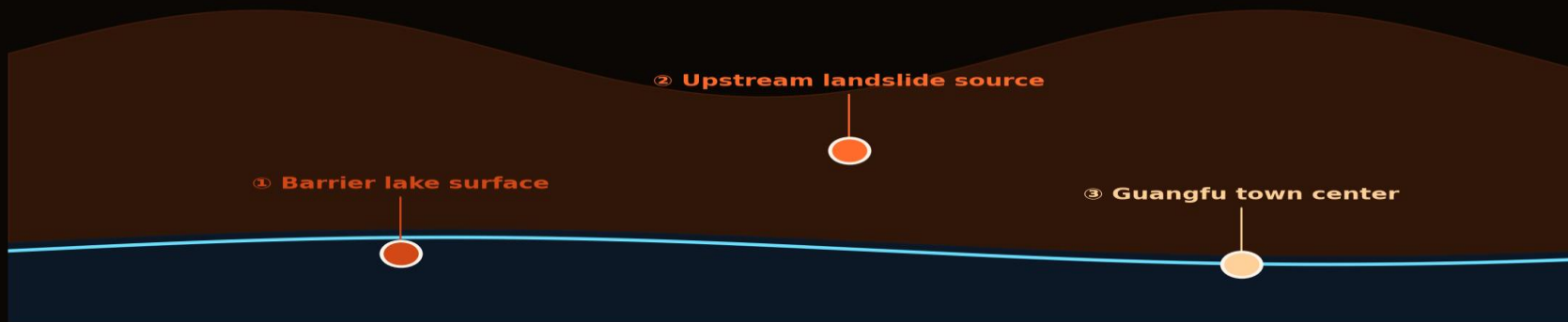
Every Sentinel-2 L2A product ships with a pre-computed **TCI** as `item.assets["visual"]`.

August and September in Hualien are peak monsoon — ~60% of scenes are cloudy.

Cheap preview before expensive band math.

Matai'an Spectral Detective

Lab 1: drop 3 points, extract spectra from Pre-event scene



1 Plot spectra

6-band vector for each site

2 Find the gap

Which band most separates forest from paddy?

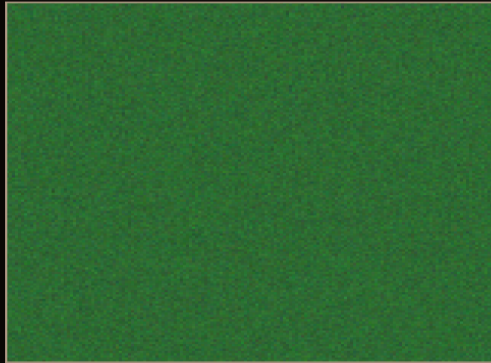
3 Predict the lake

Forecast the lake-surface spectrum you'll see in Lab 2

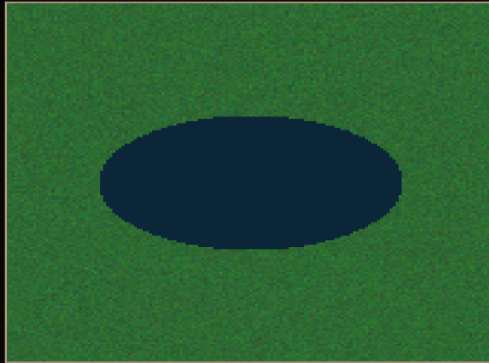
New Water Body — the Lake Signal

Case A — New Water Body (Pre → Mid NIR drop)

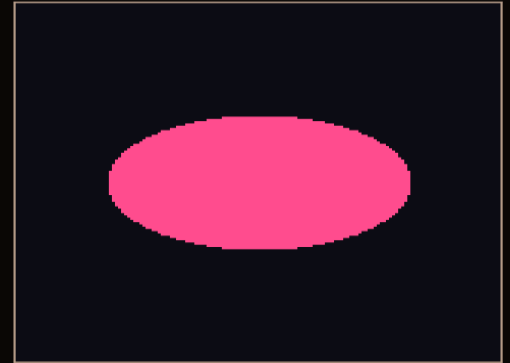
Pre (NIR high)



Mid (NIR collapse)



NDWI mask → lake polygon



THE PHYSICS

Liquid water absorbs NIR almost completely. A pixel that was forest becoming water is the cleanest signal in RS.

ARIA DETECTION RULE

```
pre_NIR > 0.30    AND    mid_NIR < 0.05    →  
lake_polygon  
(Pre→Mid)
```

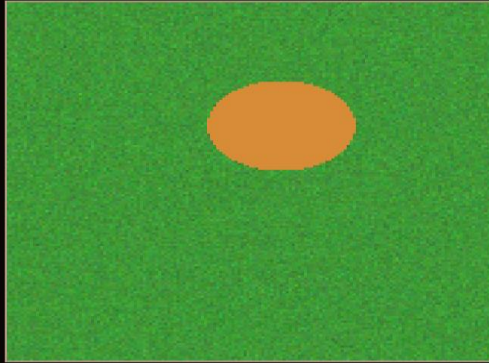
Landslide Source — the Spectral Shift

Case B — Landslide Source (Pre → Post NIR drop + SWIR spike)

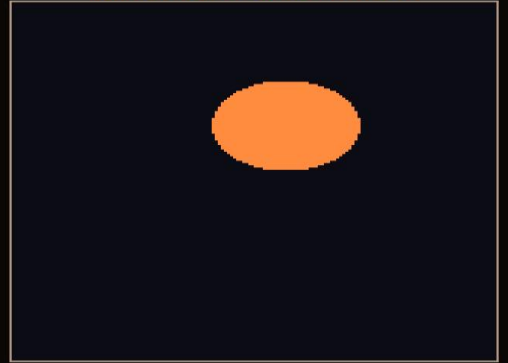
Pre SWIR comp.



Post SWIR comp.



Landslide source mask



THE PHYSICS

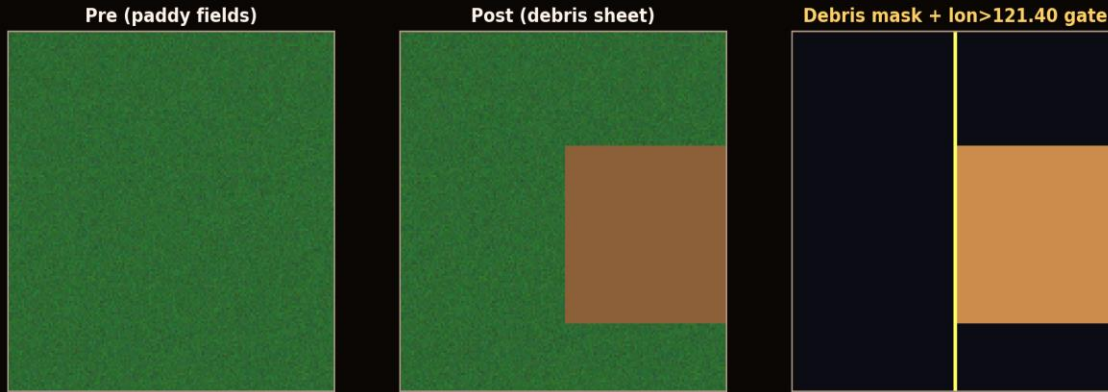
A landslide replaces vegetation (high NIR) with bare soil or rock (high SWIR). Bright orange scar in a SWIR composite.

ARIA DETECTION RULE

```
(pre_NIR - post_NIR) > 0.15  AND  post_SWIR > 0.25  AND  
pre_NIR > 0.25  
(Pre→Post)
```

Debris Flow Footprint + NBR

Case C — Debris Flow Footprint (Pre → Post NDVI drop + BSI spike + downstream gate)



NBR (general)

$$(NIR - SWIR) / (NIR + SWIR)$$

high → healthy

low/neg → burned

SWIR sees through smoke that blocks visible light.

ARIA DEBRIS RULE

$(pre_NDVI - post_NDVI) > 0.25$ AND $post_BSI > 0.15$ AND $lon > 121.40$ (downstream gate)

Same sensor, same algorithm family. Swap the AOI and the date — everything else stays.

Case Study Gallery

2023

**Canadian
wildfires**

—

SWIR through smoke

2022

Pakistan floods

—

Massive NIR blackout

2024

**Hualien 0403
earthquake**

—

*SWIR scar mapping · same
region*

2025

**Matai'an barrier
lake**

—

Our case this week

Same COGs, same STAC API, same Python code. *Only the AOI and the date change.*

Ground Truth — Esri Sentinel-2 Explorer

Independent Verification — Esri Sentinel-2 Explorer

SCENE METADATA

Scene ID	S2C_MSIL2A_20250911T02...
Satellite	Sentinel-2C
Acquired	Sep 11, 2025
Cloud Cover	14%
Relative Orbit	46
Renderer	Natural Color

WHAT TO LOOK FOR

-  **Barrier lake**
Turquoise water body in the valley center
-  **Landslide scar**
Bright bare rock face upstream of the lake
-  **Cloud wall**
Upper half of scene limits the detection window

livingatlas.arcgis.com/sentinel2explorer | Sep 11, 2025 | Natural Color

WHAT THIS PROVES

The Sep 11, 2025 scene from Esri's viewer is the exact same scene our Python notebook selects — S2C, orbit 46, 14% cloud. The barrier lake is clearly visible.

WHY IT MATTERS

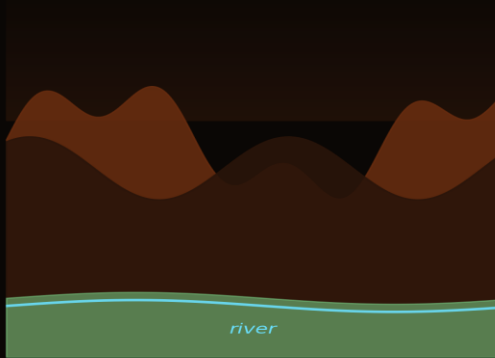
Our STAC API + stackstac pipeline retrieves the same data as commercial GIS platforms. Students can cross-check any result in a browser — no code needed.

The Matai'an Three-Act Audit

The Matai'an Three Acts

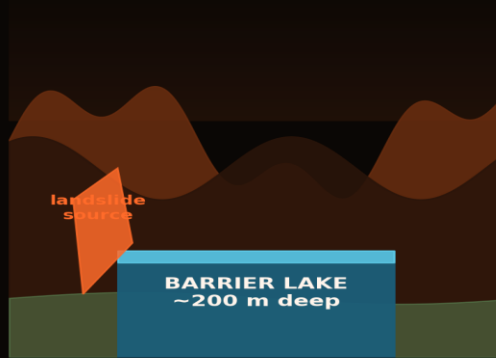
Act 1 — Pre (Jun 2025)

Forested valley, intact river



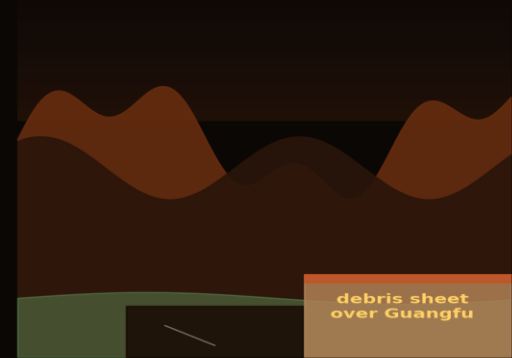
Act 2 — Mid (Aug 2025)

~200 m deep barrier lake



Act 3 — Post (Oct 2025)

Lake drained; debris sheet over Guangfu



C1 Lake polygon

NIR drop + low blue
(Pre → Mid)

C2 Landslide source

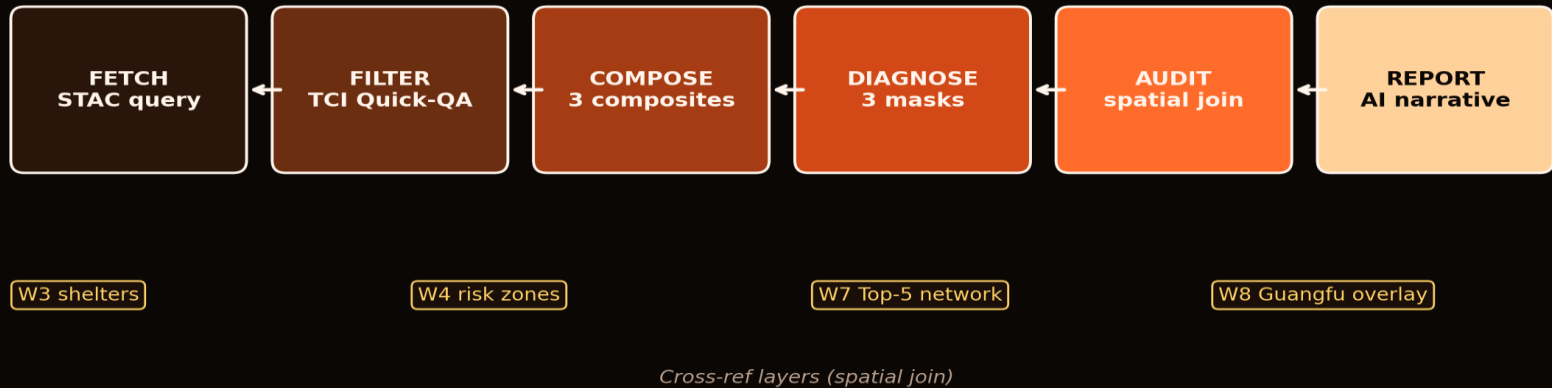
NIR drop + SWIR spike
(Pre → Post)

C3 Debris footprint

NDVI drop + BSI spike
+ downstream gate

ARIA v5.0 — The Matai'an Three-Act Auditor

ARIA v5.0 — Matai'an Three-Act Auditor



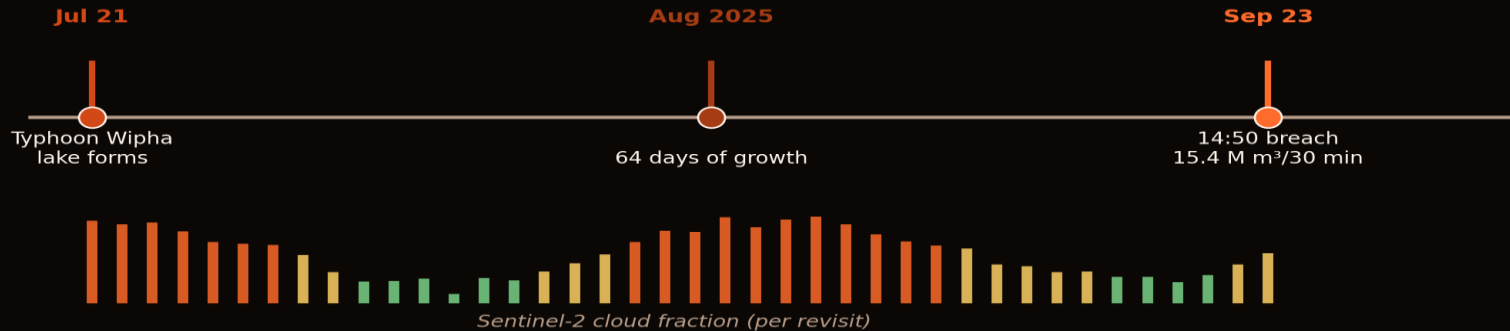
VERSION CHAIN

v4.0 W7 network → **v5.0 W8 Matai'an Auditor** → v6.0 W9 change metrics → v7.0 W10 SAR penetration

Bridge to Week 10 — When Optical Fails

Only ~4 truly clean optical windows in 64 days

→ Week 10: SAR (Sentinel-1) sees through clouds



NEXT WEEK · W9

Quantitative change metrics · NDVI, NDWI, NBR · confusion matrix

TWO WEEKS · W10

SAR · Sentinel-1 · sees through clouds, day or night

WEEK 8 REFLECTION

Pixels are measurements of energy, not colors.

YOU ARE NOW

A Spectral Intelligence Officer — able to diagnose disasters from 786 km above.

AI ADVISOR

Any LLM you prefer: ChatGPT · Gemini · Claude · local model. Cell [20] of the Demo notebook shows a vendor-neutral API pattern.

NEXT WEEK · W9

Quantitative Change Detection — NDVI, NDWI, NBR, and the confusion matrix that tells you if your detection is actually correct.

A model predicts the disaster. A satellite witnesses it.

Never ignore the eyewitness — especially when the eyewitness has been taking pictures of the same lake for 64 days straight.